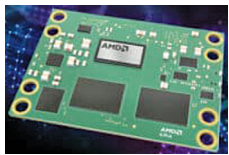


electric motor systems, public transportation infrastructure, healthcare equipment, and electric vehicle charging setups. In collaboration with the KD240 drives



starter kit, a ready-to-go motor control development platform, it simplifies the transition from development to production, reducing time to market.

AMD

<https://www.amd.com/en.html>

32-bit MCU with built-in security module

Microchip Technology has launched a series of PIC32CZ CA 32-bit microcontrollers. Powered by a 300MHz Arm Cortex-M7 processor, these microcontrollers are equipped with an embedded hardware security module (HSM) and offer a range of Flash memory and connectivity options



for enhanced flexibility. The HSM serves as a comprehensive security solution, functioning as a secure enclave.

Additionally, the microcontrollers support factory provisioning for devices requiring advanced security measures, ensuring customers receive pre-configured, secure devices ready for deployment. These MCUs offer versatile connectivity options, including USART/UART, I2C, SPI, CAN FD, USB, and Gigabit Ethernet, with support for audio video bridging and Precision Time Protocol for Gigabit Ethernet, adhering to IEEE 1588 standards.

Microchip Technology

<https://www.microchip.com>

Embedded systems debugging tool

Percepio has introduced the Tracealyzer SDK that delivers extensive

observability tools, comprising RTOS awareness and API tracing, enabling processor vendors and software professionals to offer advanced observability. Essential for debugging and profiling embedded, edge, and IoT systems, the tool's enhanced features



facilitate rapid development and debugging, guaranteeing software dependability and peak performance. While supporting well-known RTOS like FreeRTOS, Zephyr, and Azure RTOS ThreadX, and adaptable to any C/C++ software, Percepio oversees its integration. The SDK's advantages extend to silicon vendors, RTOS, and middleware sectors, enhancing design success with superior tooling. It offers in-depth insights into RTOS, applications, and various layers, expediting customer development and production orders. Furnished with comprehensive documentation and code samples, the Tracealyzer SDK ensures easy integration with Percepio's open source TraceRecorder, which aids users in profiling CPU tasks, software timing, and memory leak detection.

Percepio

<https://percepio.com>

Raspberry Pi 5 with enhanced features

Raspberry Pi has introduced a new version of its board, Raspberry Pi 5. The Raspberry Pi 5 will be available in 4GB and 8GB models, boasting a performance over twice as fast as its predecessor. While retaining the iconic credit-card-sized form, the design has been tweaked to cater to its enhanced chipset. Where the display connector once was, a compact flexible printed circuit connector now facilitates a

single lane of PCI Express 2.0 for high-speed peripherals. Powering the Raspberry Pi 5 are three integral chips. The BCM2712, a 16-nanometre application



processor by Broadcom housing a quad-core 64-bit ARM Cortex-A76 processor

running at 2.4GHz, accompanied by 512kB per-core L2 caches and a shared 2MB L3 cache. The RP1 chip, fabricated via TSMC's 40-nanometre 40LP process, manages the Raspberry Pi RP2040 microcontroller, delivering functionalities like USB interfaces, an Ethernet controller, and more. Lastly, the Renesas DA9091 "Gilmour" PMIC efficiently supports the operations of the BCM2712 and RP1.

Raspberry Pi

<https://www.raspberrypi.com>

MISCELLANEOUS

PLC mini control units making control effortlessly visible

Display Visions has launched a programmable logic controller (PLC) mini control unit family, a leap forward in the PLC systems world. The Core central control unit prioritises not just control but clear communication. It boasts a vibrant 7.1cm (2.8-inch) IPS



touchscreen display for immediate, intuitive data visualisation.

It can connect with up to 50

satellite displays, turning each into a monitoring setup and personal viewing hub. The Core features built-in Ethernet, RS-485 interface, and Wi-Fi for streamlined communication. Data parameters on the Core and satellites update in real time. The company provides PLC Designer, a free Windows tool with a graphical WYSIWYG edi-